UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

ELE1204: Digital Electronics and Logic Design

Lecture 03 (BSCS)

Topic/Theme, e.g., *Combinational Logic Design*

Ms. Olivia Nakayima
(MSc, BSc. CCNA, JNCIA, JNCIS)
Department of Computing & Technology
Faculty of Engineering, Design & Technology

Tue 11th February 2025

Lecture Objectives and Learning outcomes

The Objectives of this lecture are :

- To understand the combinational circuits
- To analyse them and their applications

By the end of this lecture, students should be able to:

- Define combinational circuits
- Fully explain the adders, multiplexers, encoders and decoders

NOTE:: SMART = **S**pecific, **M**easurable, **A**chievable, **R**elevant, **T**imebound

Lecture Overview

- Introduction and characteristics
- Analysis of basic combinational circuits
- Adders (Half, Full), Subtractors
- Multiplexers (MUX) and Demultiplexers (DEMUX)
- Encoders and Decoders, Code Converters Parity Generators and Checkers

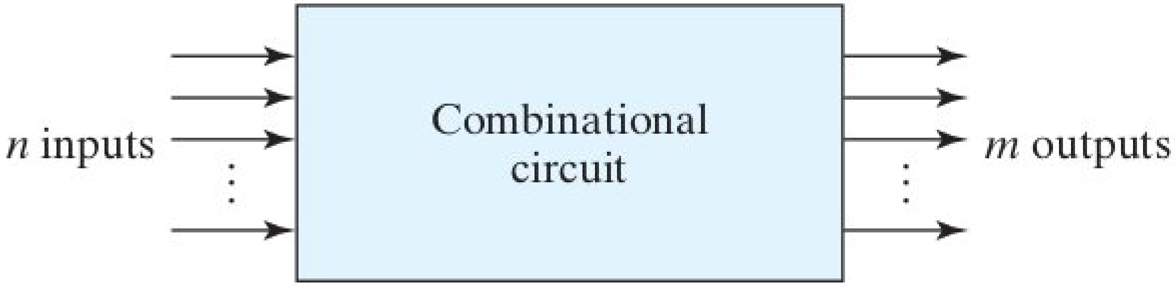
Introduction to Digital circuit Simulation Tools (Logisim, multisim)

http://www.cburch.com/logisim/

Introduction and characteristics

Introduction

Combinational logic circuits are digital circuits in which the **output depends only on the present input values** and not on previous inputs (no memory). These circuits use **logic gates** (AND, OR, NOT, XOR, etc.) to perform operations



Characteristics of Combinational Circuits.

- **Output depends only on the current input values** (No feedback or memory).
- **Faster operation** as there is no need to store past values.
- **Built using basic logic gates** such as AND, OR, NOT, XOR, NAND, and NOR.
- **Used in arithmetic and data-processing applications** like adders, multiplexers, and decoders.

Analysis of basic combinational circuits

Steps to Analyze a Combinational Circuit

- **Identify the inputs and outputs**.
- **Create the truth table** that defines the behavior of the circuit.
- **Derive the Boolean expressions** for each output.
- **Simplify the Boolean expressions** using Boolean algebra or Karnaugh Maps (K-Maps).
- **Draw the logic circuit diagram** based on the simplified expression.

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill | Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: https://ucu.ac.ug
Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda

Arithmetic Circuits

Adders

An adder is a combinational circuit used for **binary addition.**

Half Adder (HA)

Adds two **single-bit** binary numbers (**X and Y**).

Outputs:

Sum (S): XOR of inputs → S=X⊕Y

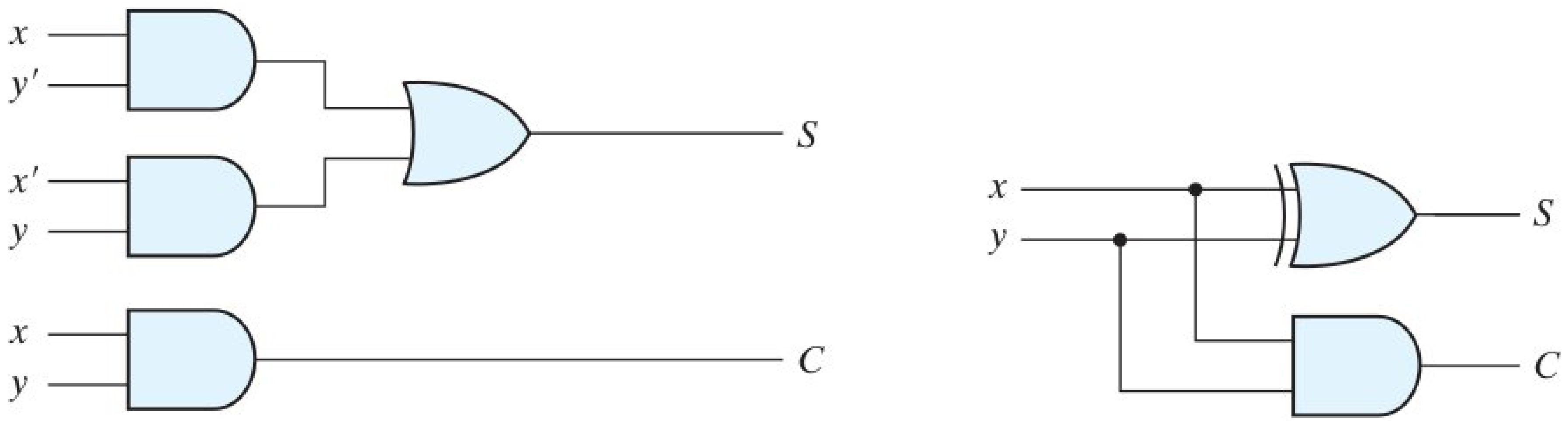
- **Carry (C)**: AND of inputs → C=X·Y

$$S = x'y + xy'$$
$$C = xy$$

X	Y	Sum(S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Arithmetic Circuits

Half Adders



(a) $S = xy' + x'y$
$C = xy$

(b) $S = x \oplus y$
$C = xy$

Arithmetic Circuits

X	Y	Z_in	Sum (S)	C_out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Full Adder

Adds **three inputs**: **X, Y, and Carry-in (Z_in)**.
Outputs:

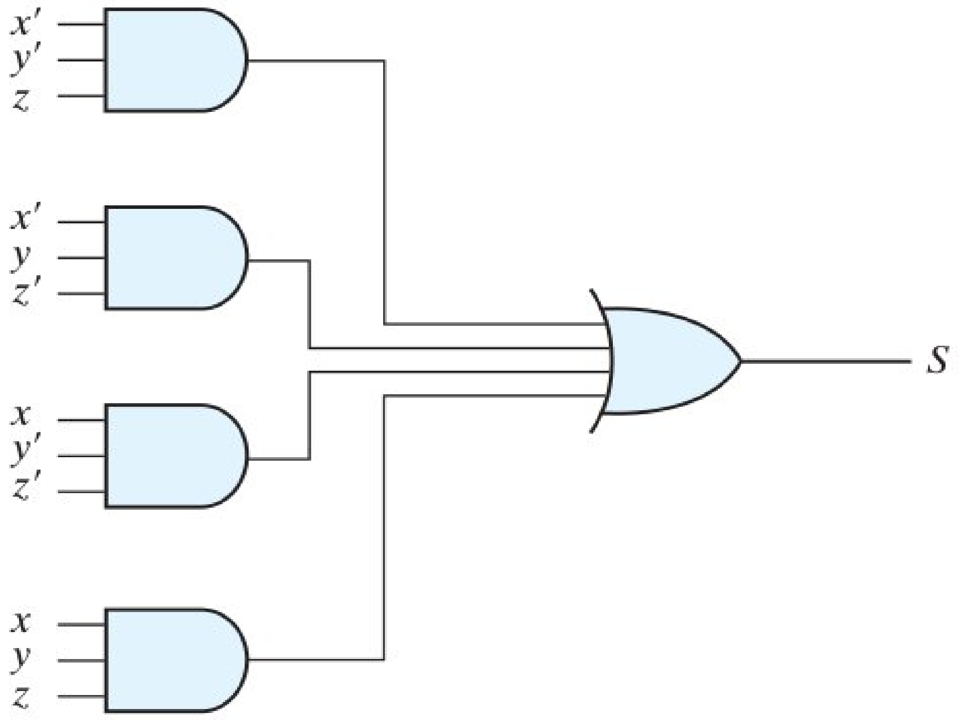
- The S output is equal to 1 when only one input is equal to 1 or when all three inputs are equal to 1. The C output has a carry of 1 if two or three inputs are equal to 1

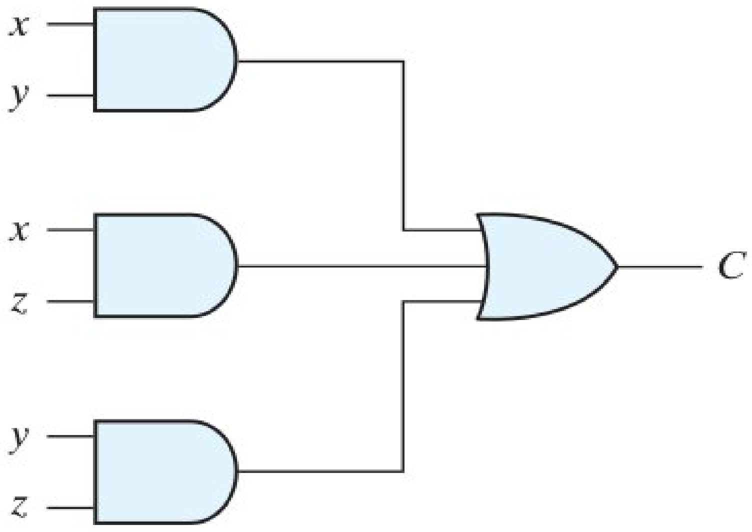
Arithmetic Circuits

Full Adder

$$S = x'y'z + x'yz' + xy'z' + xyz$$

$$C = xy + xz + yz$$





Arithmetic Circuits

Subtractors

A **subtractor** is a combinational circuit used for **binary subtraction**.

Assignment

Read about the half and full subtractors and design their circuits from their truth tables.

Encoders and Decoders

Encoder

- Converts **2^n inputs** into **n outputs**.
- **Example: 8-to-3 Encoder** converts **8 input lines into a 3-bit binary code**.

Decoder

Converts **n inputs into 2^n outputs**.

Example: 3-to-8 Decoder takes **3-bit input and activates one of 8 output lines**.

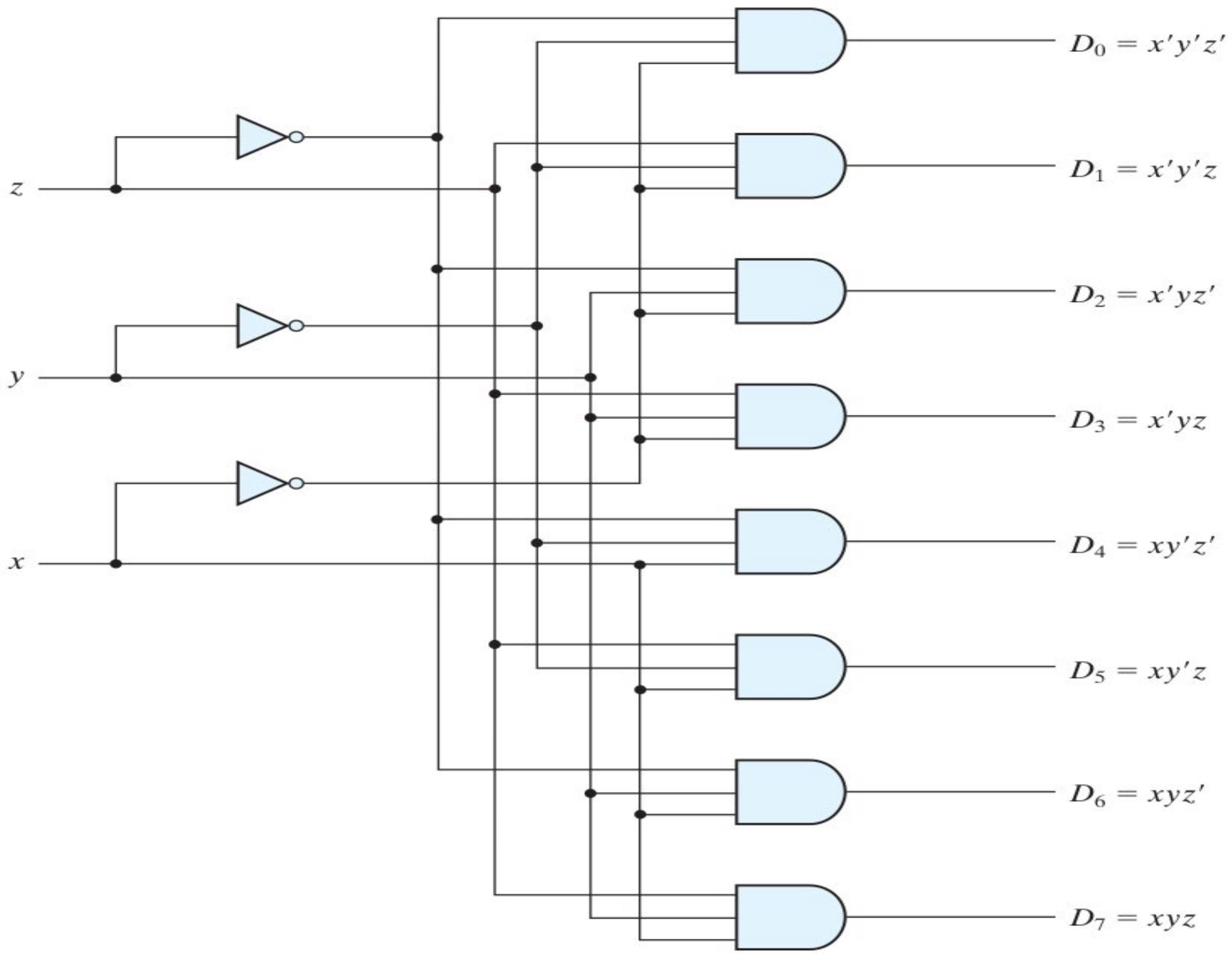
Encoders and Decoders

Decoder

- A decoder is a combinational circuit that converts binary information n input lines to a maximum of **2^n outputs**.
- If the n -bit coded information has unused combinations, the decoder may have fewer than **2^n outputs**.
- A particular application of this decoder is binary-to-octal conversion

https://www.youtube.com/watch?v=a3wh7xV5PBU

Example: 3-to-8 Decoder takes **3-bit input and activates one of 8 output lines.**



Encoders and Decoders

Decoder

- The output whose value is equal to 1 represents the minterm equivalent of the binary number currently available in the input lines.
- Decoders include one or more enable inputs to control the circuit operation.

Applications

Code conversion, address decoding in memory

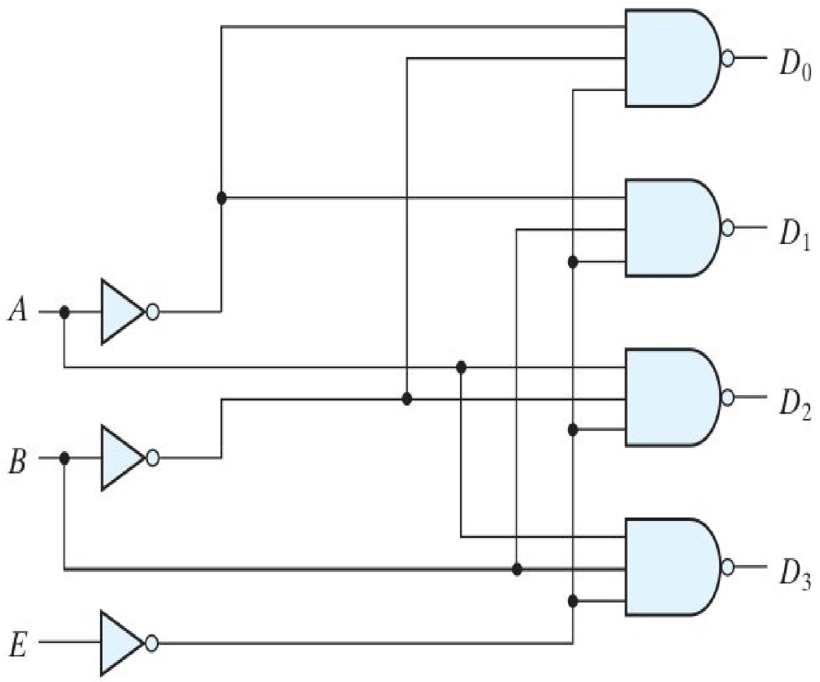
Truth Table of a Three-to-Eight-Line Decoder

Inputs			Outputs							
x	y	z	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Encoders and Decoders

Two-to-four-line decoder with enable input

- Some decoders are constructed with NAND gates.
- Decoders include one or more enable inputs to control the circuit operation.
- The decoder is enabled when E is equal to 0 (i.e., active-low enable).
- Output can be equal to 0 at any given time.
- The output whose value is equal to 0 represents the minterm selected by inputs A and B.
- A decoder with enable input can function as a demultiplexer



(a) Logic diagram

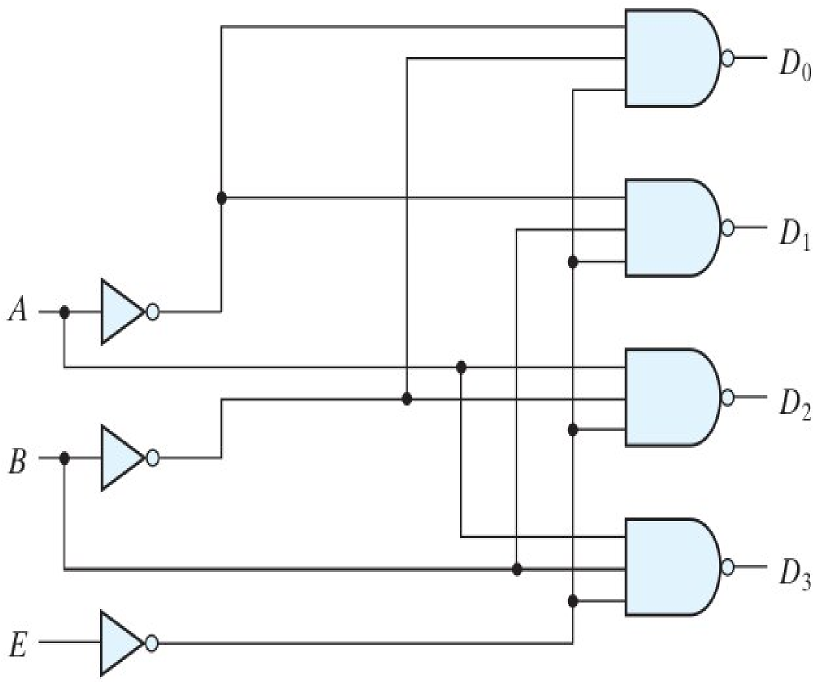
E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	1

(b) Truth table

Encoders and Decoders

Two-to-four-line decoder with enable input

- The circuit is disabled when E is equal to 1.
- When the circuit is disabled, none of the outputs are equal to 0 and none of the minterms are selected.



(a) Logic diagram

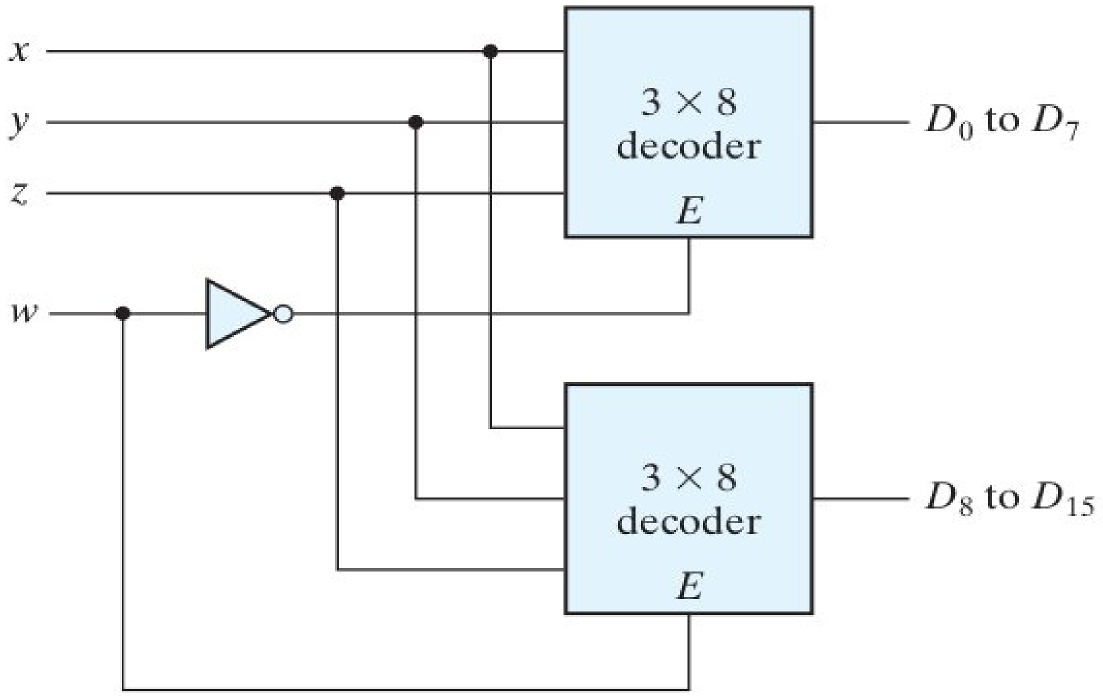
E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	1

(b) Truth table

Encoders and Decoders

4 * 16 decoder constructed with two 3 * 8 decoders

- Decoders with enable inputs can be connected together to form a larger decoder circuit.
- When **w =0**, the top decoder is enabled and the other is disabled.
- The bottom decoder outputs are all 0's, and the top eight outputs generate minterms **0000 to 0111**.
- When **w=1**, the enable conditions are reversed: The bottom decoder outputs generate minterms **1000 to 1111**, while the outputs of the top decoder are all 0's



Encoders and Decoders

Encoder

- performs the inverse operation of a decoder.
- It is assumed that only one input has a value of 1 at any given time.
- The encoder can be implemented with OR gates whose inputs are determined directly from the truth table.
- The encoder can be implemented with three OR gates.

Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Encoders and Decoders

Encoder

- Output z is equal to 1 when the input octal digit is 1, 3, 5, or 7. Output y is 1 for octal digits 2, 3, 6, or 7, and output x is 1 for digits 4, 5, 6, or 7.
- These conditions can be expressed by the following Boolean output func

$$z = D_1 + D_3 + D_5 + D_7$$
$$y = D_2 + D_3 + D_6 + D_7$$
$$x = D_4 + D_5 + D_6 + D_7$$

Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Encoders and Decoders

Encoder

- limitation is that only one input can be active at any given time.
- If two inputs are active simultaneously, the output produces an undefined combination.
- To resolve this ambiguity, encoder circuits must establish an input priority to ensure that only one input is encoded.
- *https://www.youtube.com/watch?v=NWiPVMDh7GE*

Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Encoders and Decoders

Priority Encoder

- A priority encoder is an encoder circuit that includes the priority function.
- The operation of the priority encoder is such that if two or more inputs are equal to 1 at the same time, the input having the highest priority will take precedence.
- A valid bit indicator is set to 1 when one or more inputs are equal to 1.
- If all inputs are 0, there is no valid input and V is equal to 0.
- The other two outputs are not inspected when V equals 0 and are specified as don't-care conditions.

Encoders and Decoders

Priority Encoder

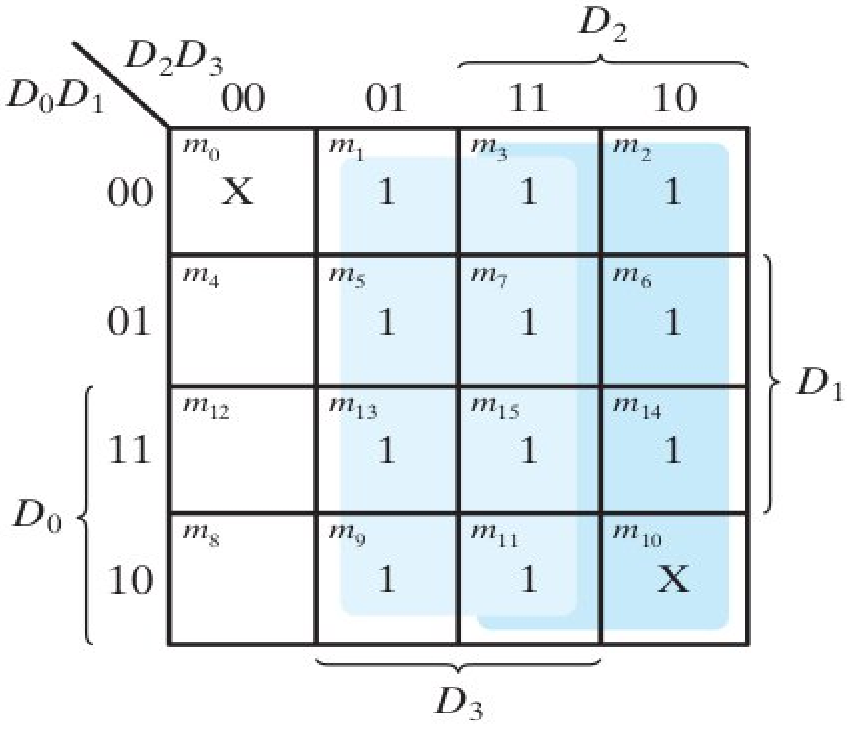
- the higher the subscript number, the higher the priority of the input.

Truth Table of a Priority Encoder

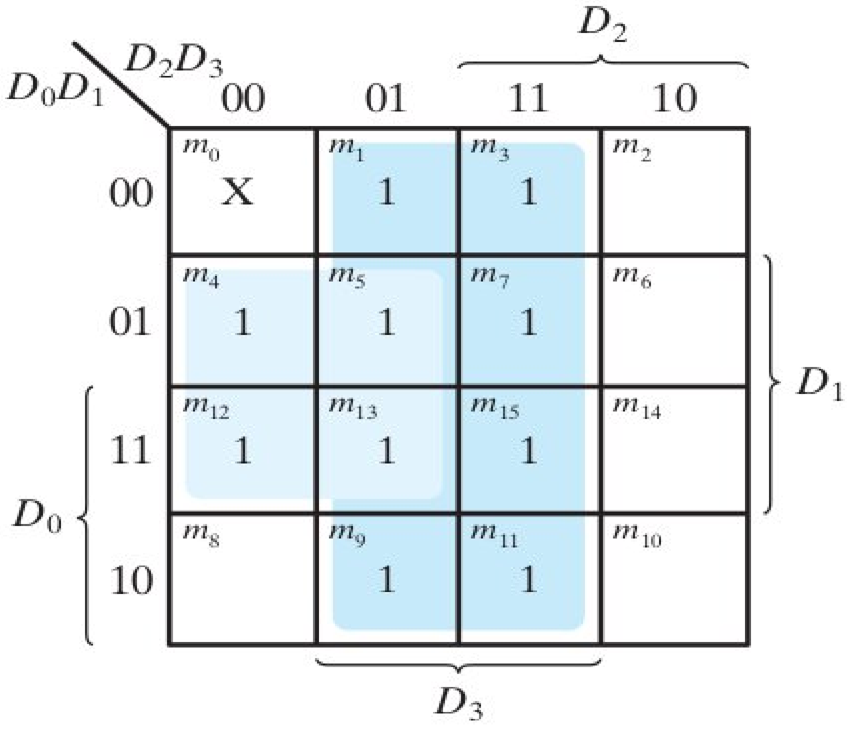
Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

Encoders and Decoders

Priority Encoder



$x = D_2 + D_3$

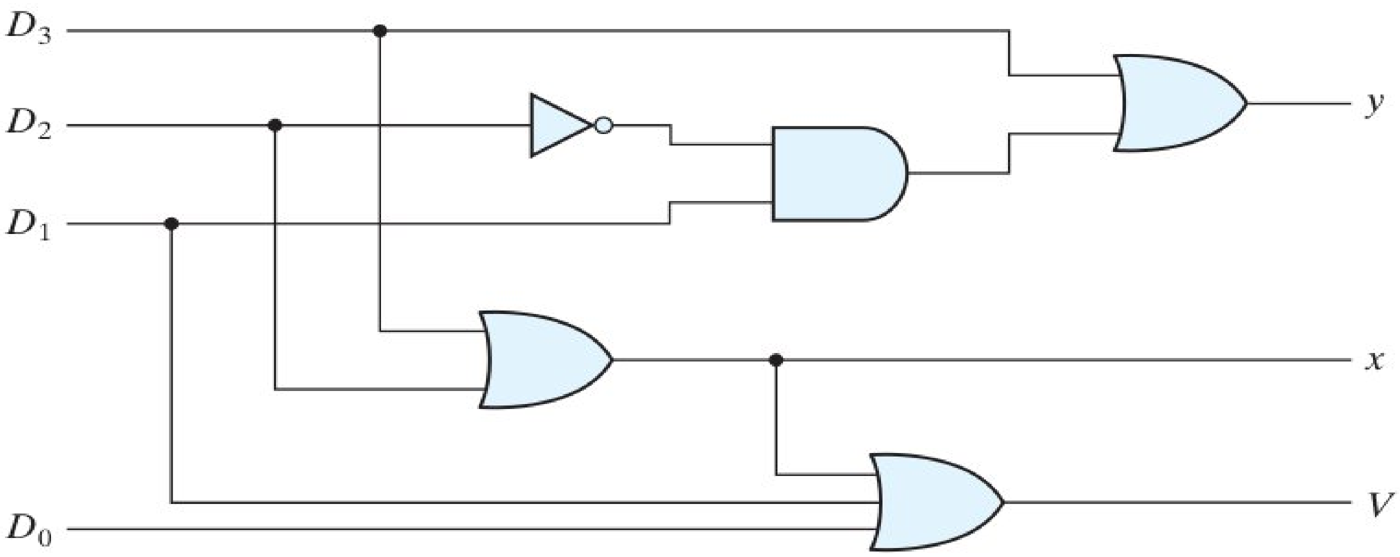


$y = D_3 + D_1D'_2$

Encoders and Decoders

Priority Encoder

$$x = D_2 + D_3$$
$$y = D_3 + D_1 D_2'$$
$$V = D_0 + D_1 + D_2 + D_3$$



Thank You

Uganda Christian University
P.O. Box 4 Mukono, Uganda
Tel: 256-312-350800
https://ucu.ac.ug/ Email: info@ucu.ac.ug.
@ugandachristianuniversity @UCUniversity
@UgandaChristianUniversity

Department of Computing & Technology
FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
Tel: +256 (0) 312 350 863 | WhatsApp: +256 (0) 708 114 300
@ucucomputeng @ucu_ComputEng
https://cse.ucu.ac.ug/ Email: dct-info@ucu.ac.ug